

Lesson 8 - Check How Good the Model Is

READ ME FIRST (theory). Then open the practice notebook.

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Lesson 8 - Check How Good the Model Is

Training is not enough. We must measure the model. The theory, then `practice.ipynb`.

Key words

Word	Simple meaning
accuracy	how many predictions are correct overall
precision	of our “Yes” guesses, how many were right
recall	of all real “Yes”, how many we caught
confusion matrix	a table of correct vs wrong, per class
overfitting	great on train, bad on test

Word	Simple meaning
underfitting	bad on train and test

1. Always test on unseen data

We check the model on the **test** set (data it never saw). This gives an honest score.

2. Accuracy

Accuracy = correct predictions / all predictions.

```
from sklearn.metrics import accuracy_score
accuracy_score(y_test, y_pred)
```

3. Accuracy can lie (very important)

If 95% of customers do NOT churn, a lazy model that always says “no churn” gets 95% accuracy - but catches ZERO churners. So accuracy alone is not enough when one class is rare.

This is why we also use precision and recall.

4. Precision and Recall

- **Precision** = of all the cases we predicted “Yes”, how many were really Yes.
- **Recall** = of all the real “Yes” cases, how many we actually caught.

Example (catching churners): - High precision = when we say “will churn”, we are usually right. - High recall = we catch most of the real churners.

F1 combines precision and recall into one number.

5. Confusion matrix

A small table showing correct and wrong predictions for each class:

```
[[true No,  wrong No ],
 [wrong Yes, true Yes]]
```

It shows exactly where the model makes mistakes.

6. Overfitting vs underfitting

- **Overfitting** = 99% on train, 60% on test. The model memorized; it fails on new data.
 - **Underfitting** = 60% on train AND 60% on test. The model is too simple.
 - **Good** = high on both, and close to each other.
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Quick summary

- Test on **unseen** data for an honest score.
- **Accuracy** can lie when one class is rare - also use **precision** and **recall**.
- **Confusion matrix** shows where mistakes happen.
- **Overfitting** = good train, bad test. **Underfitting** = bad on both.

Now do this

Open `practice.ipynb`, measure the model, and do the assignment.